Revision · Graphs, gradient and intercept

Cambridge insert: reading a straight line — and reading a story off one.

LESSON 94–96 | 3 × 40 MIN | ALGEBRA 8 – REVISION | STAGE 9 · 10.1–10.4

LESSON CLOCK

6' | 12' | 10' | 8' | 4'

Sketch three lines from memory	6 min
Gradient and intercept	12 min
Parallel and perpendicular	10 min
Real-life graphs	8 min
Homework	4 min

LEARNING OBJECTIVES

- Find the gradient and y-intercept of a straight line from its equation or its graph.
- Draw a line from $y = mx + c$ without a full table of values.
- Recognise parallel and perpendicular lines from their gradients.
- Interpret real-life graphs, including distance–time and conversion graphs.

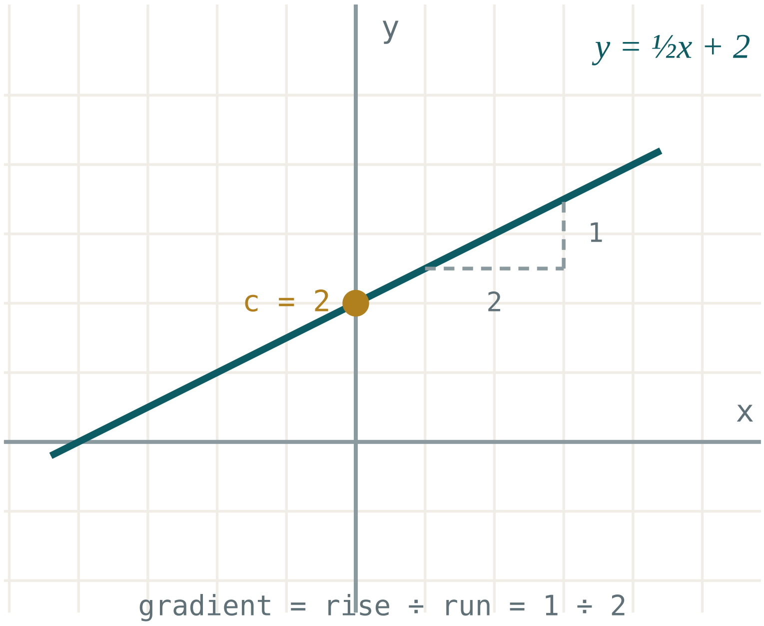
TEXTBOOK

National	Annual revision
Cambridge	Learner's Book pp. 214–234
Workbook	Workbook 10.1–10.4

01 Explanation

y = mx + c, read straight off

> $y = mx + c$ — m is the gradient, c is where the line cuts the y-axis



The gradient is rise ÷ run; c is the height at which the line crosses the y-axis.

$y = 2x - 3$ climbs 2 for every 1 across and passes through $(0, -3)$. That is enough to draw it: mark $(0, -3)$, then step 1 right and 2 up, and join.

GRADIENT FROM TWO POINTS

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Take the differences in the **same order** top and bottom, or the sign comes out wrong.

A positive gradient rises left to right, a negative one falls, and a gradient of 0 is a horizontal line $y = c$. A vertical line $x = k$ has **no** gradient — the run is zero and you cannot divide by it.

Parallel and perpendicular

RELATIONSHIP	GRADIENTS	EXAMPLE
parallel	$m_1 = m_2$	$y = 3x + 1 \parallel y = 3x - 4$
perpendicular	$m_1 \cdot m_2 = -1$	$y = 3x + 1 \perp y = -\tfrac{1}{3}x + 2$

To find the perpendicular gradient: turn the fraction upside down and change its sign. For $m = \dfrac{2}{5}$ the perpendicular gradient is $-\dfrac{5}{2}$.

An equation given as $2x + 3y = 12$ hides its gradient. Rearrange into $y = mx + c$ form first:

$$3y = -2x + 12 \Longrightarrow y = -\frac{2}{3}x + 4$$

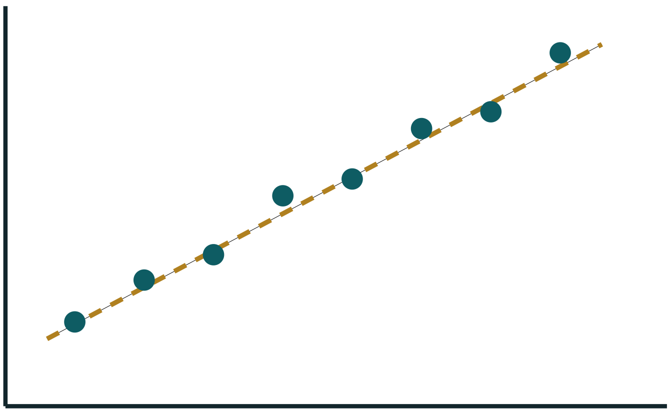
Gradient $-\frac{2}{3}$, intercept 4.

Graphs that tell a story

On a **distance–time** graph the gradient *is* the speed:

- a steep line — moving fast;
- a gentle line — moving slowly;
- a horizontal line — stopped;
- a line coming back down — returning towards the start.

Always read the axes before the shape. The same rising line means “getting faster” on a speed–time graph but “moving at a steady speed” on a distance–time graph.



positive correlation

A scatter graph shows the relationship between two quantities; a line of best fit summarises the trend.

On a **scatter graph** the trend can be positive, negative or absent. A line of best fit passes through the middle of the cloud with roughly as many points above as below — and it is only trustworthy **inside** the range of the data.

> CORRELATION IS NOT CAUSE
>
> Ice-cream sales and swimming accidents rise together. Neither causes the other; hot weather causes both.

02 Worked examples

EXAMPLE 1 **Find the gradient and y-intercept of $4x + 2y = 10$, and state a parallel line.**

1. $2y = -4x + 10$

   Get the y term alone.

2. $y = -2x + 5$

   Divide everything by 2.

3. $m = -2, c = 5$

4. $y = -2x + 1$

   Any line with the same gradient and a different intercept is parallel.

ANSWER

$m = -2, c = 5$

EXAMPLE 2 **Find the equation of the line through $(1, 5)$ and $(4, 14)$.**

1. $m = \dfrac{14 - 5}{4 - 1} = \dfrac{9}{3} = 3$

   Same order top and bottom.

2. $y = 3x + c$

3. $5 = 3 \cdot 1 + c$

   Substitute one of the points.

4. $c = 2$

5. $y = 3x + 2$

   Check with $(4, 14)$: $12 + 2 = 14$ ✓

ANSWER

$y = 3x + 2$

EXAMPLE 3

A car travels 120 km in 1.5 h, stops for 0.5 h, then returns in 2 h. Describe the distance–time graph.

1. $0 \to 1.5\ h$

   A straight rise to 120 km; gradient $80\ km/h$.

2. $1.5 \to 2\ h$

   Horizontal at 120 km — stopped.

3. $2 \to 4\ h$

   Falls back to 0; gradient $-60\ km/h$, so a speed of $60\ km/h$ homeward.

4. Total journey 4 h, 240 km travelled.

   The **displacement** at the end is 0.

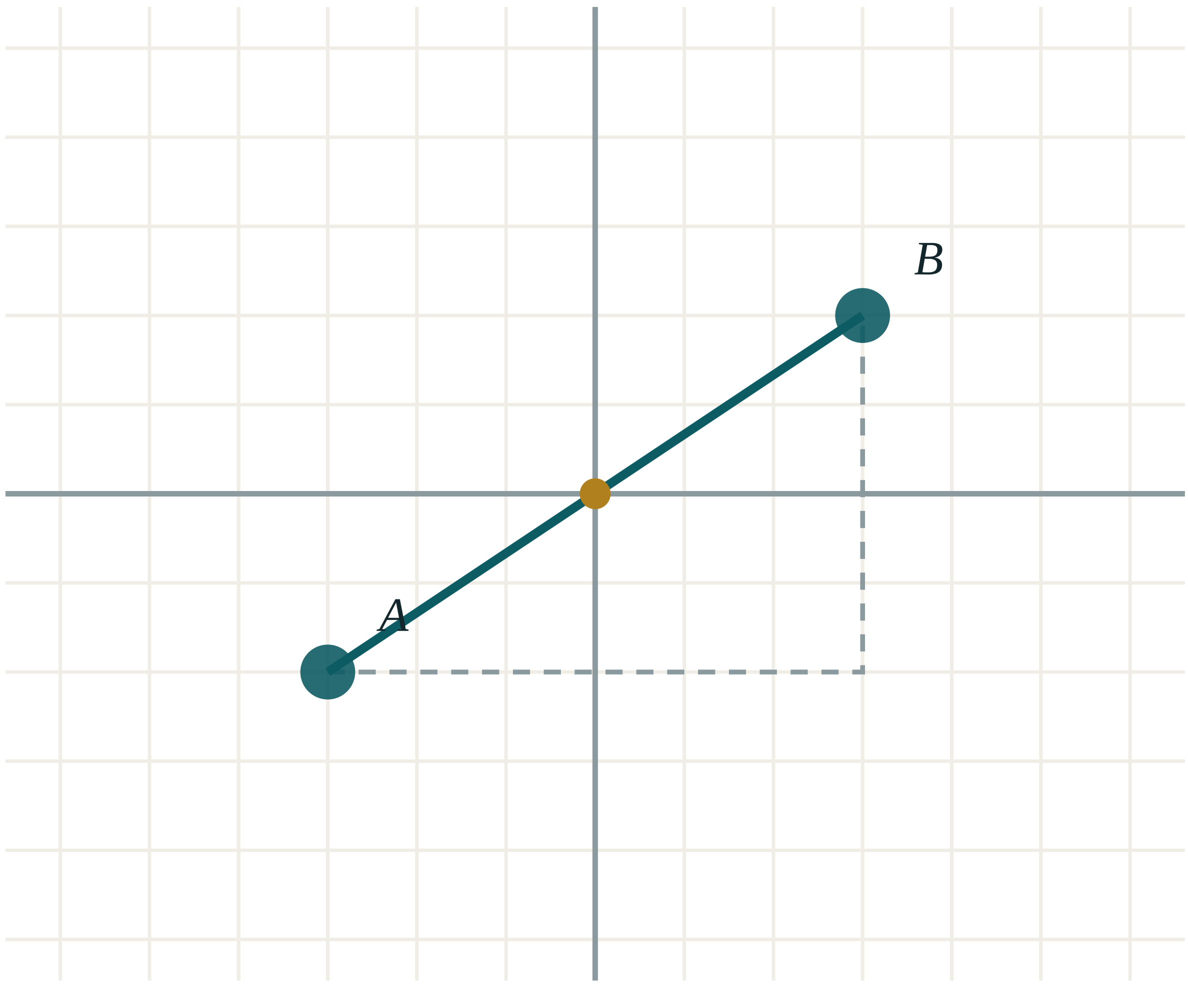
ANSWER

Rise (80 km/h), flat (stopped), fall (60 km/h)

03 Interactive model

Drag the two points and read the gradient off the change in x and the change in y.

Gradient between two points



A (-3, -2) | B (3, 2) | $x_2 - x_1$ 6 | $y_2 - y_1$ 4 | distance AB 7.21 | midpoint M (0, 0)

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Gradient (slope)	Burchak koeffitsienti	Угловой коэффициент
y-intercept	Ordinatalar o‘qi bilan kesishuv	Точка пересечения с осью Oy
x-intercept	Abssissalar o‘qi bilan kesishuv	Точка пересечения с осью Ox
Rise over run	Ko‘tarilish / siljish	Приращение по y / по x
Parallel lines	Parallel to‘g‘ri chiziqlar	Параллельные прямые
Perpendicular lines	Perpendikulyar to‘g‘ri chiziqlar	Перпендикулярные прямые
Linear equation	Chiziqli tenglama	Линейное уравнение
Distance–time graph	Masofa–vaqt grafigi	График «путь — время»
Rate of change	O‘zgarish tezligi	Скорость изменения
Conversion graph	Almashtirish grafigi	График перевода единиц

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The gradient of $y = 5 - 3x$ is:

- 5
- 3
- -3
- -5

A line perpendicular to $y = 4x - 1$ has gradient:

- 4
- -4
- $-\frac{1}{4}$
- $\frac{1}{4}$

The line through $(0, 3)$ and $(2, 9)$ is:

- $y = 3x + 3$
- $y = 2x + 3$
- $y = 6x$
- $y = 3x + 2$

On a distance–time graph a horizontal segment means:

- constant speed
- stopped
- going backwards
- accelerating

Rearranged, $3x - y = 7$ is:

- $y = 3x - 7$
- $y = 7 - 3x$
- $y = 3x + 7$
- $y = -3x - 7$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *State the gradient and intercept of* $y = 4x - 1$.

   ANSWER $m = 4, c = -1$

2. *State the gradient and intercept of* $y = -x + 6$.

   ANSWER $m = -1, c = 6$

3. *Find the gradient through* $(0, 0)$ *and* $(2, 6)$.

   ANSWER 3

4. *Does* $(2, 7)$ *lie on* $y = 3x + 1$?

   ANSWER Yes — $6 + 1 = 7$

5. *Where does* $y = 2x - 8$ *cut the* y*-axis?*

   ANSWER $(0, -8)$

6. *Write a line parallel to* $y = 5x + 2$.

   ANSWER $y = 5x - 3$ (any different intercept)

7. *What is the gradient of* $y = 7$?

   ANSWER 0

Medium · the standard question

7 PROBLEMS

1. *Rearrange $2x + y = 9$ into $y = mx + c$.*

   ANSWER $y = -2x + 9$

2. *Find the gradient through (1, 2) and (5, 14).*

   ANSWER 3

3. *Find the equation through (0, −4) with gradient 2.*

   ANSWER $y = 2x - 4$

4. *Where does $y = 3x - 12$ cut the x-axis?*

   ANSWER (4, 0)

5. *Write a line perpendicular to $y = 2x + 1$ through the origin.*

   ANSWER $y = -\frac{1}{2}x$

6. *A walker covers 6 km in 1.5 h. Find the gradient of the distance–time graph.*

   ANSWER 4 km/h

7. *Rearrange $5x - 2y = 8$ and state the gradient.*

   ANSWER $y = \frac{5}{2}x - 4,\ m = \frac{5}{2}$

Hard · stretch and reason

7 PROBLEMS

1. *Find the equation through $(2, 1)$ and $(6, -7)$.*

   ANSWER $y = -2x + 5$

2. *Are $3x + 2y = 6$ and $y = \frac{2}{3}x - 1$ perpendicular?*

   ANSWER Yes — gradients $-\frac{3}{2}$ and $\frac{2}{3}$ multiply to -1.

3. *Find where $y = 2x + 1$ meets $y = 8 - x$.*

   ANSWER $(\frac{7}{3}, \frac{17}{3})$

4. *A line has gradient 3 and passes through $(-2, 1)$. Find its x-intercept.*

   ANSWER $y = 3x + 7$, so $(-\frac{7}{3}, 0)$

5. *The line $y = mx + 4$ passes through $(3, -2)$. Find m.*

   ANSWER $m = -2$

6. *Show that $(1, 3)$, $(3, 7)$ and $(6, 13)$ are collinear.*

   ANSWER All three pairs give gradient 2, and each point satisfies $y = 2x + 1$.

7. *A taxi charges 5000 so‘m plus 2000 so‘m per km. Write the cost as a function of distance and interpret m and c.*

   ANSWER $C = 2000d + 5000$ — m is the price per km, c the fixed hire charge.

07 Homework

Homework — 6 problems

Cambridge Stage 9, Unit 10. Draw axes from -6 to 6 unless told otherwise.

1. *State the gradient and intercept of $y = -3x + 5$, and draw it.*
2. *Rearrange $4x + 2y = 10$ and state its gradient.*
3. *Find the equation of the line through $(0, 2)$ and $(4, 10)$.*

4. *Write a line parallel to, and a line perpendicular to,* $y = \frac{1}{2}x + 3$.

5. *A cyclist rides 24 km in 2 h, rests 0.5 h, then rides 12 km in 1 h. Sketch the distance–time graph.*

6. In one sentence, explain why correlation does not prove cause.